Lab Report: Text classification with neural networks

1 Introduction

In this report we compare several models on standard benchmarks. Each experiment was repeated 3 times with different random seeds. All experiments were run on a single GPU with 24 GB of memory.

2 Results

Figure 1 shows the test accuracy of Baseline by dataset.

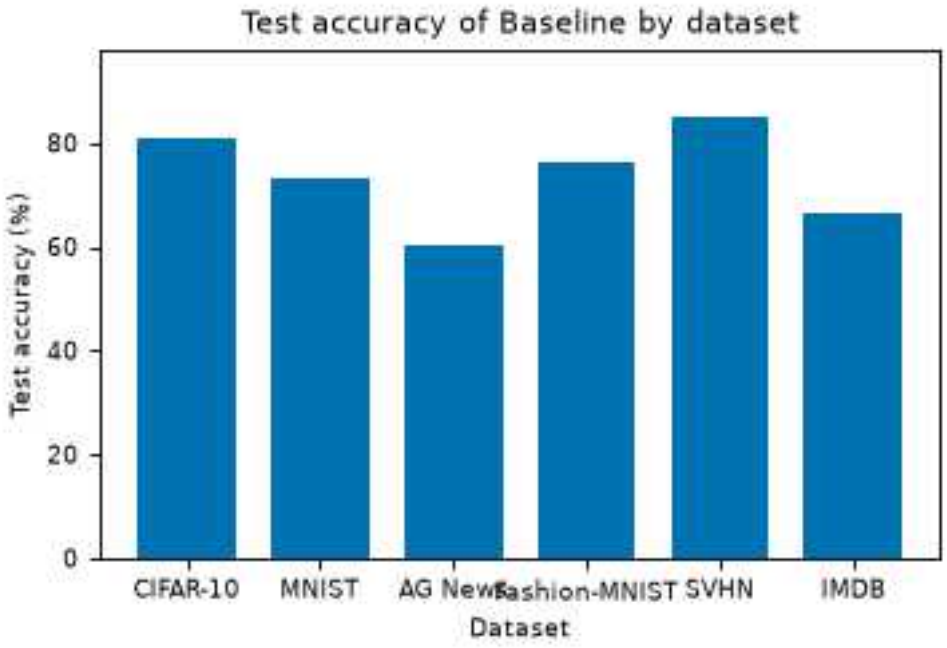


Figure 1: Test accuracy of Baseline by dataset for the models in this experiment.

Baseline scores 57.5 points higher on SVHN than on Fashion-MNIST (Figure 1). As shown in Figure 1, Baseline achieves 65.5% test accuracy on SVHN.

Figure 2 shows the training loss.

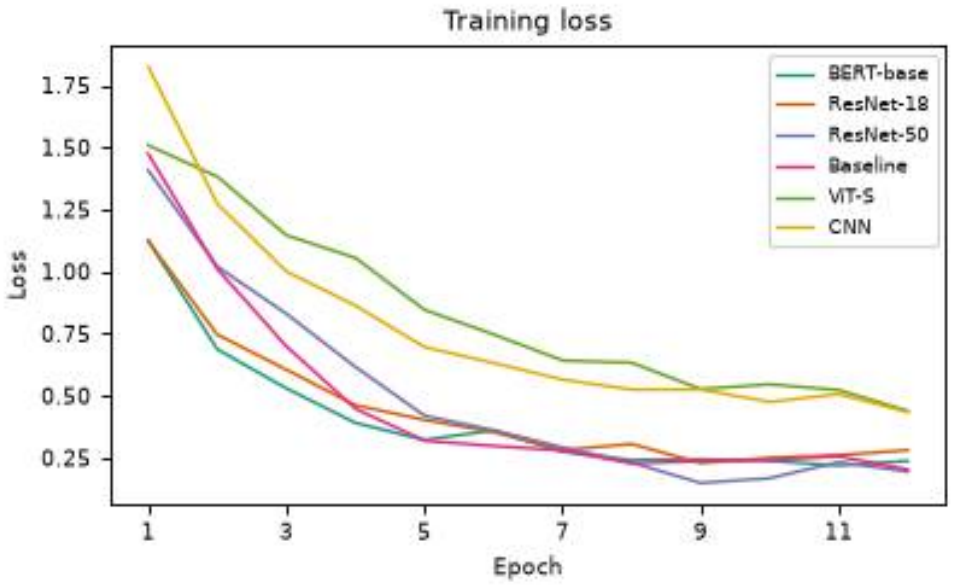


Figure 2: Training loss for the models in this experiment.

As shown in Figure 2, at epoch 9, the loss of ResNet-50 is 0.186. At epoch 8, the loss of Baseline is 0.136 (Figure 2). At epoch 12, the loss of Baseline is 0.203 (Figure 2).

3 Discussion

SVM reaches 54.2% accuracy on Tiny-ImageNet. The differences between the models are discussed in the next section.